



Report on Electricity Tariffs in Assam and Impact of Proposed 1000 MWp Solar Project

November 2023

Asian Development Bank

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List of Acronyms:

Acronym	Definition
ABR	Average Billing Rate
ACOS / ACS	Average Cost of Supply
AERC	Assam Electricity Regulatory Commission
A&G	Administrative and General
APDCL	Assam Power Distribution Company Limited
APGCL	Assam Power Generation Corporation Limited
APM	Administered Pricing Mechanism
APPC	Average Power Purchase Cost
APR	Annual Performance Review
BN	Billion
BU	Billion Units of Electricity
CAGR	Compounded Annual Growth Rate
COD	Commissioning Date
CSGS	Central Sector Generation Stations
CUF	Capacity Utilization Factor
DPR	Detailed Project Report
DRE	Distributed Renewable Energy
DSM	Deviation Settlement Mechanism
FY	Financial Year
GW	Giga Watt
GWh	Gigawatt hour
INR	Indian Rupee
kWh	Kilowatt hour
LCOE	Levelized Cost of Electricity
MNRE	Ministry of New and Renewable Energy, Govt. of India
MN	Million
MU	Million Units of Electricity
MW	Mega Watt
MWp	Mega Watt Peak
NER	Northeastern Region
NSDP	Net State Domestic Product
O&M	Operations and Maintenance
PPA	Power Purchase Agreement
PPC	Power Procurement Cost
R&M	Renovation & Modernization
RES	Renewable Energy Sources
ROE	Return on Equity
RPO	Renewable Purchase Obligation
Rs.	Indian Rupee
SHP	Small Hydro Plant
SLDC	State Load Despatch Center
SLM	Straight Line Method
USD	United States Dollar
WACC	Weighted Average Cost of Power
YoY	Year on Year

1 Introduction

Over the past decade, the North-Eastern Region of India, with a special emphasis on Assam, has witnessed remarkable economic development. Assam's Per Capita Net State Domestic Product (NSDP) has grown at a Compound Annual Growth Rate (CAGR) of approximately 5% from Financial Year (FY) 12 to FY 23, surpassing the national average¹. Simultaneously, the demand for electricity in Assam has steadily increased at a CAGR of around 6%, exceeding national averages in the last five years. Assam's strategic location near neighboring countries like Bangladesh, Bhutan, Nepal, China, and Myanmar presents unique opportunities for economic, strategic, and cultural collaborations in the vast Asia-Pacific region.

Despite these successes, Assam faces energy security challenges. Historically, the State has relied on gas and hydro-based power generation, facing challenges related to gas availability and price fluctuations. Further the State is dependent on electricity supply from outside the State (due to limited intra State supply capacity) for meeting the rising electricity demand. The electricity procurement costs in the State have risen sharply in recent years due to increase in fuel prices which have put significant strain on the financials of Assam Power Distribution Company Limited as it continues to cater to the entire State's electricity supply. The Government of Assam is committed to reduce dependence on electricity supply from outside the State (as transmission constraints also make it difficult to increase supply capacity) and promote renewable energy capacity addition from within the State. The Government of Assam has introduced initiatives like the Assam Renewable Energy (RE) Policy in 2022 with a target of adding 1,200 Mega-Watt (MW) of renewable energy capacity. Additionally, schemes like "Mukhya Mantri Souro Shakti Prokolpo" aim to install 1000 MW of solar projects on government-owned land, signaling a strong shift towards a greener energy portfolio.

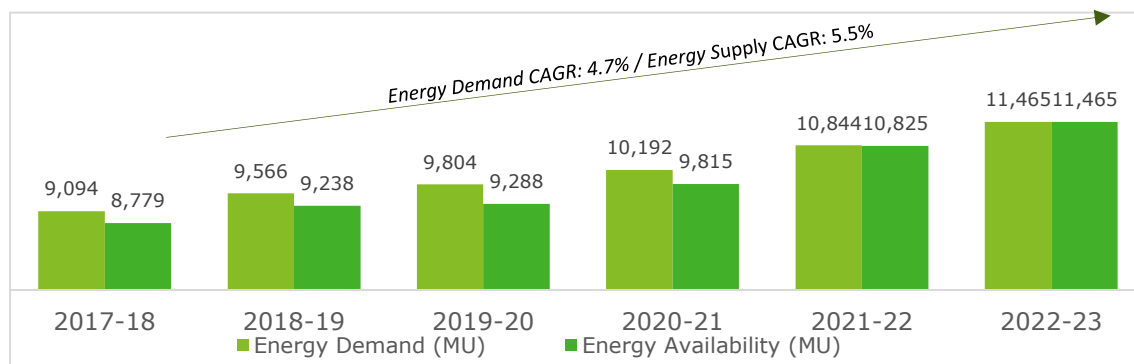
¹ Source: Ministry of Statistics and Program Implementation, Government of India (MOSPI)

2 Power Sector Profile in Assam

2.1 Energy Demand Growth

Power Demand in Assam has witnessed a rapid growth of CAGR of ~4.7% between FY18 to FY23 increasing from 9,094 MUs to ~11,465 MUs respectively, outperforming the entire Northeastern Region demand growth of ~3.3% CAGR during the same period. The State's energy supply which was in deficit of ~3.4% in FY18 has now achieved parity with demand in FY23 through mainly short-term power purchases which have come at higher costs.

Figure 1: Energy Requirement vs Supply in Assam (Million Units / Giga Watt-Hours)²:

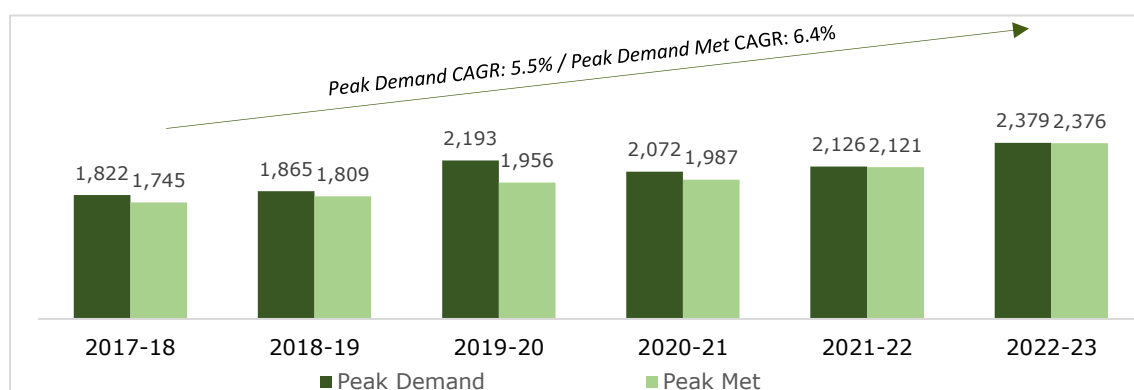


In FY 24, Energy requirement has already reached ~5,731 MUs from April to August 2023 (first 5 months of the financial year).

2.2 Peak Demand and Peak Demand Met

The peak energy demand of the State has grown at a higher rate of 5.5% CAGR rising from 1,822 MW in FY18 to 2,379 MW in FY23³. Similarly, to meet this rapid growth, supply capacity has also grown at a CAGR of ~6.4% to meet the historical deficits and reach parity with demand.

Figure 2: Peak Demand vs Met (Assam)



In FY 24, Peak Demand as on August 2023 has already reached 2,535 MW.

² CEA Monthly Power Supply Position Reports (Energy)

³ CEA Monthly Power Supply Position Reports (Peak)

2.3 Installed Capacity and Share

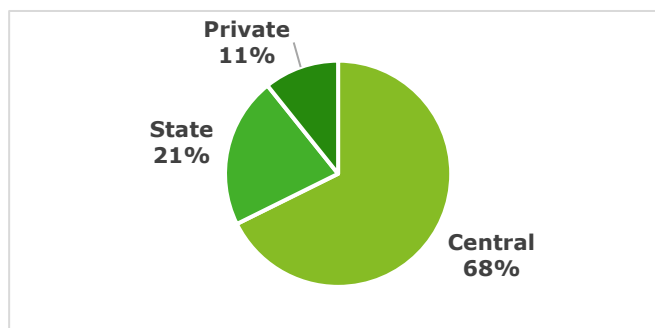
The Installed Capacity in Assam has grown by 168 MW during the period FY 2019 – FY 2023⁴. The solar capacity has grown nearly five times during the same period. On the supply side, Coal and Natural Gas dominate the installed capacity of the State with a total share of 68% followed by Hydro (18%) and Solar (12%). With the peak demand reaching ~2,500 MW in FY24 (till August) the State has been meeting its demand from short term power sources as well as Central Allocations.

Table 1: Installed Capacity of Assam (MW)

Resource-wise Capacity	FY 19	FY 20	FY 21	FY 22	FY 23	% Contribution (FY23)
Coal	750	750	750	750	750	38%
Natural Gas	644	644	645	620	600	30%
Hydro	350	350	350	350	350	18%
Solar	41	43	69	148	248	12%
Other Renewable Energy Sources (RES)	40	42	42	45	45	2%
Total	1,825	1,829	1,856	1,913	1,993	100%

In terms of capacity ownership, Central allocations contribute ~68% of the State's available capacity (in MW terms) followed by State's own share of 21% and remaining by private developers at 11%. Assam Power Generation Corporation Ltd (State Genco) has a total electricity generating capacity of ~422 MW and is dominated by gas plants (309 MW). The private share is mostly in Solar Capacity with a share of around ~203 MW of capacity.

Figure 3: Installed Capacity share w.r.t Ownership



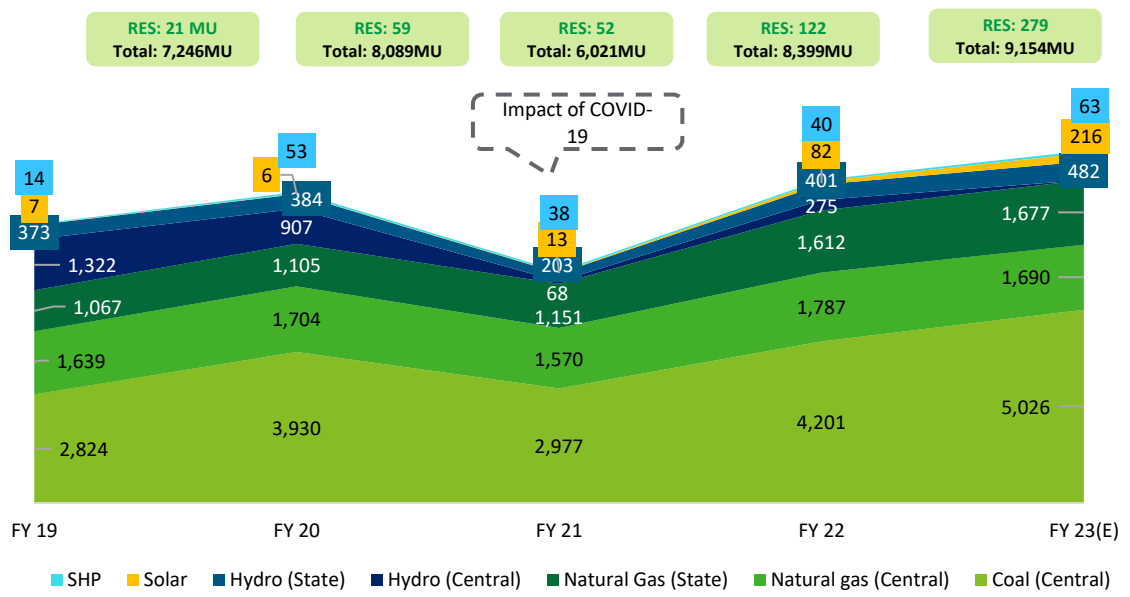
2.4 Energy Generation in Assam

The energy generation in Assam has grown from 7,246 MUs to 9,154 MUs during the period FY 2019-2024. This in spite of a decline during FY 2021 due to impact of Covid-19 but returned back to normal from FY 2022 onwards. The fossil fuels such as Coal and natural gas contributes around 55% and 37% respectively to the overall generation mix. The renewable energy sources (RES) and hydro contributes to only 8% in energy terms.

⁴ Source: CEA and APDCL data

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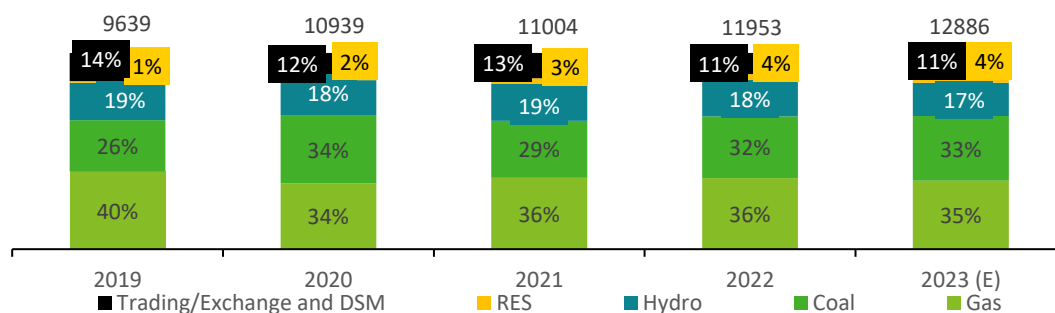
Figure 4: Energy Generation in Assam (MUs)



3 APDCL's Power Purchase Mix

To cater to state's electricity demand, Assam has allocations in the central sector generating stations (CSGS) and also relies on the direct procurement through other sources like traders, bilateral agreements and power exchanges.

Figure 5: Energy Purchase Mix of APDCL (MU)

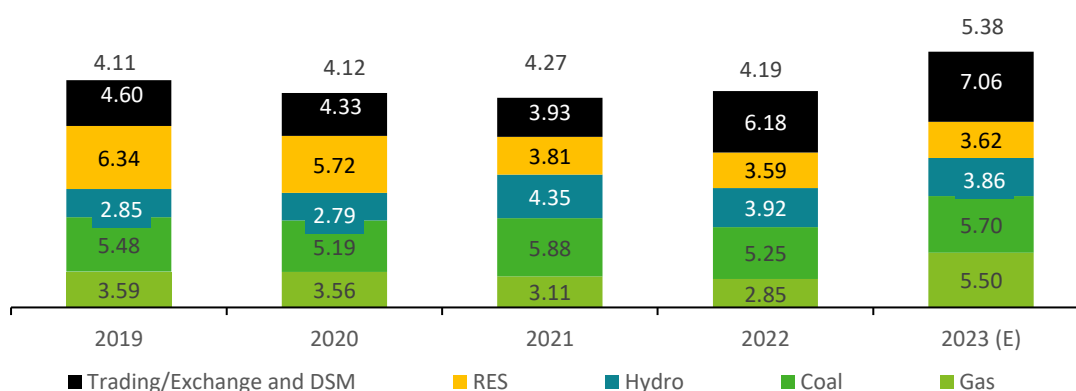


The share of coal in the power purchase portfolio has risen from 26% in FY19 to 33% in FY23(E) due to constraints in capacity expansion of gas plants (availability of gas been challenging) and limited increase in Renewable Energy Sources (RES) based capacities during the same period.

3.1 APDCL's Power Procurement Costs

The power purchase costs from gas and trading activities has increased significantly in FY23 which has led to the power purchase costs rising to nearly Rs 5.4 / KWh from historical levels of FY19 to FY22⁵.

Figure 6: Average Power Purchase Cost (Rs / kWh)

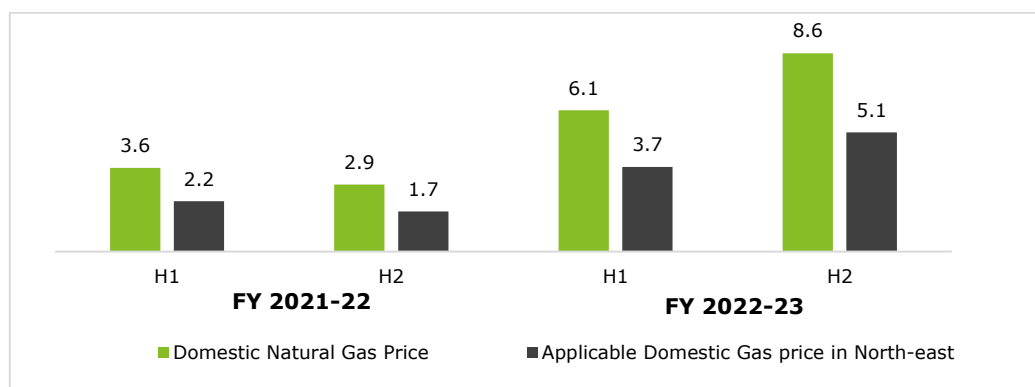


The average power procurement from gas-based plants has increased from Rs. 2.85 / kWh in FY22 to Rs. ~5.38 in FY23 due to the significant increase in natural gas prices in India. This was majorly due to the geo-political situation affecting global fuel prices which were historically linked to the domestic gas pricing formula adopted by the Govt. of India. While North-East Region gets a price discount of 40% on the natural gas pricing (under Administered Price Mechanism), the prices increased by over 200% between FY22 and FY23. From FY 24 onwards, Govt. has approved floor price of \$4/ Million British Thermal Units (MMBTu) and ceiling price of \$6.5/MMBTu for FY2024 and FY2025 for the natural gas supplied under Administered Price Mechanism with annual

⁵ Source: APR Order FY23 - APDCL

escalation of \$0.25/mmbtu on the ceiling price thereafter⁶. However, natural gas supplied from Deepwater, Ultra Deepwater and High Pressure - High Temperature areas continue to be determined by the Government under the pricing freedom mechanism. Thus, even with the floor and ceiling mechanism under APM, generating companies (gencos) such as Assam Power Generation Corporation Ltd (APGCL) which have significant gas-based capacity and source both APM and non-APM gas, would continue to face price volatility issues which would impact APDCL in the downstream value chain.

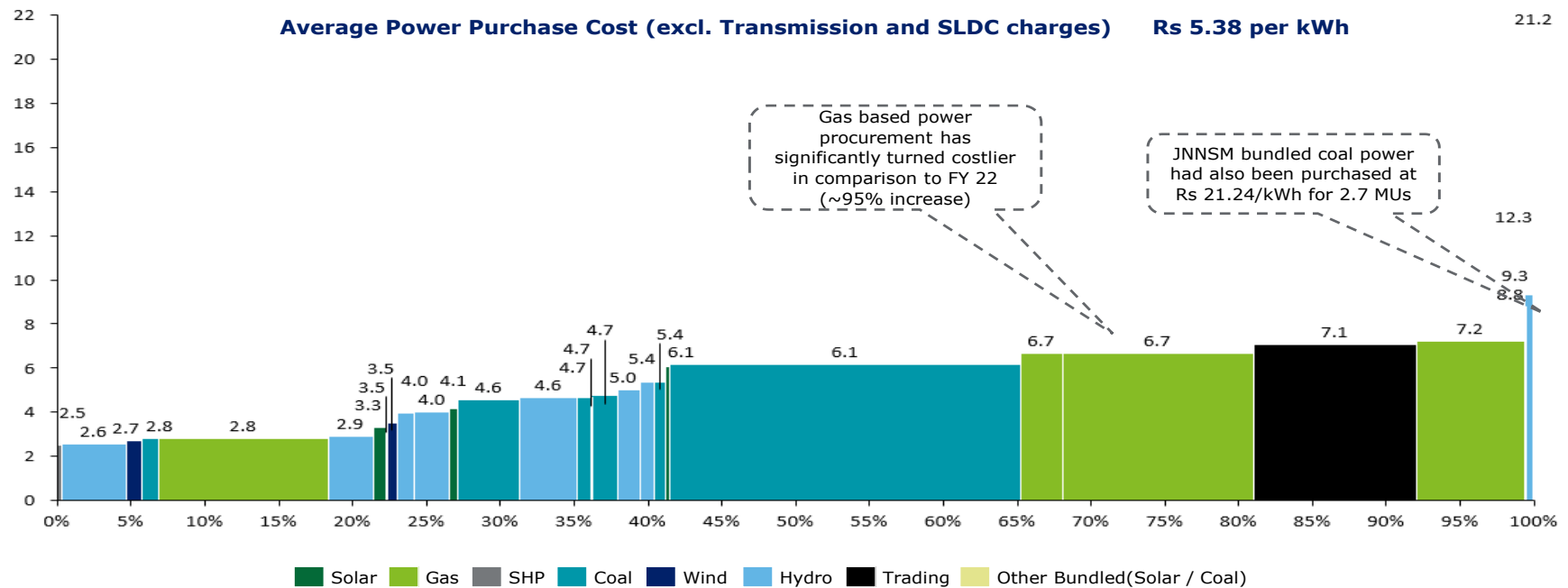
Figure 7: Half-yearly gas price comparison - FY 22 and FY 23 (USD / MMBTu)



Similarly, the power purchase cost from trading / DSM (Deviation Settlement Mechanism) segment also witnessed a significant rise from average cost of Rs. 3.93 / kWh in FY21 to Rs. 7.06 / kWh in FY23 which has been due to the surge in national power demand driving short term prices coupled with increasing fuel prices. Even coal-based power has been costing an average tariff of over Rs 5.50 / kWh which is driving the average power purchase cost for APDCL.

⁶ Source: <https://mopng.gov.in/files/natural-gas/policies-and-guidelines/notification-dt-7th-april-2023.pdf>

Figure 8: Merit Order Dispatch - FY 23 (Estimated)



As can be seen from the graph above (based on total power purchase cost for APDCL APR Order for FY23), about ~60% of the total power purchase quantum comprising of coal, gas and trading-based power is being bought at over Rs. 6 / kWh creating financial challenges for APDCL which in turn impacts the retail tariffs for the State.

In comparison the Average Power Procurement Cost (APPC) for Eastern and Northeastern States are considerably lower⁷:

⁷ Source: Tariff Orders for FY22-23 issued by respective State Electricity Regulatory Commissions

Table 2: APPC of Eastern and North-Eastern States

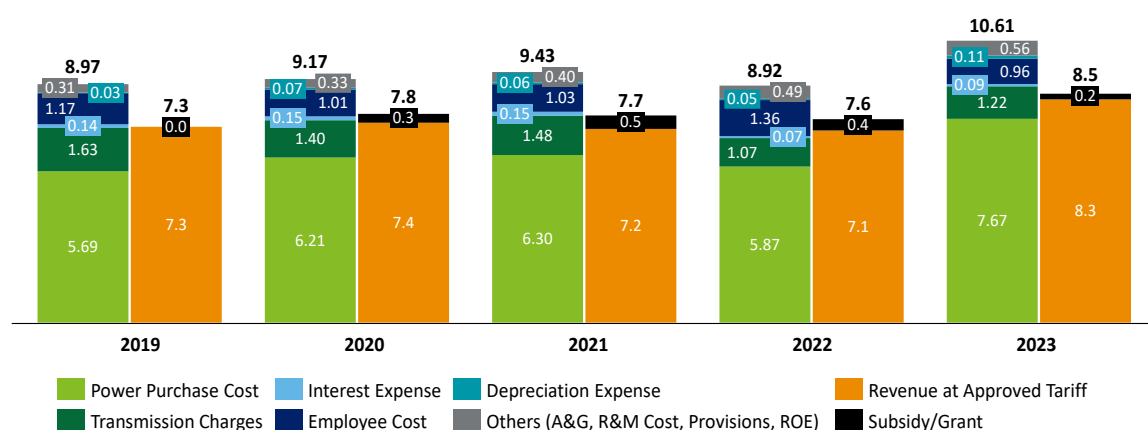
State	APPC (Rs. Per kWh)
Tripura	4.33
Meghalaya	4.19
Manipur	5.76
Jharkhand	4.41
Bihar	4.92
West Bengal	3.42

For most of the Eastern and North-Eastern States, the APPC is lower than Rs 5 / kWh. With Assam being a key NER state in terms of economic activity, the high APPC for the State is expected to be a deterrent in attracting investments as required.

3.2 Gap between Average Cost of Supply and Average Billing Rate for APDCL

During the last 5 years, the Average Cost of Supply (ACS) for APDCL has increased at a CAGR of 4%. The major components are power purchase cost (72% of ACS); Transmission charges (12% of ACS); Employee Expenses (9% of ACS) and Others (includes Interest Expenses, Depreciation, A&G, R&M, Provisions and ROE). Average Cost of Supply has increased significantly from FY 22 to FY 23 majorly attributable to increase in high-cost power purchase.

Figure 9: ACS-ABR (in INR/kWh) Gap of last 5 Years



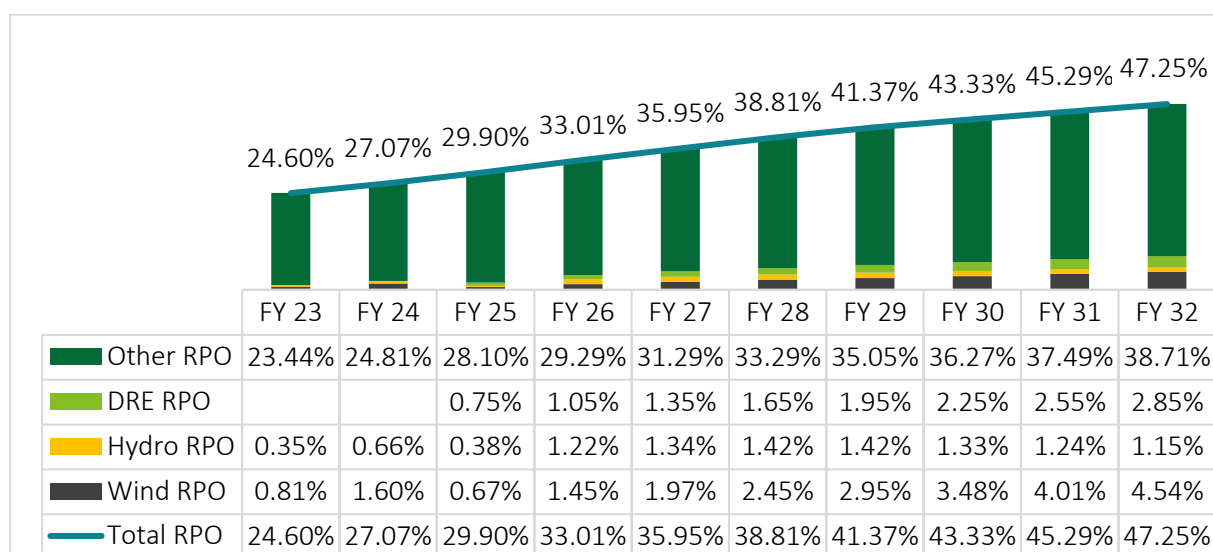
In Assam, the Average Cost of Supply (ACS) has been higher than the Average Billing Rate (ABR) over last five years as depicted in the graph above. The difference is projected to increase from Rs. 1.3 / kWh in 2022 to over Rs 2 / kWh in FY23 which may create significant headwinds for APDCL to operate sustainably.

4 Importance of Inclusion of RE in the energy supply mix for APDCL

4.1 Renewable Purchase Obligations

Considering the present power procurement portfolio of APDCL, inclusion of Renewable Energy in Assam is important to reduce the dependency on fossil fuels such as Gas and Coal based power procurement which are coming in at a very high rate in recent years. Further, the renewable purchase obligation (RPO) targets setup by Ministry of Power (MoP) vide notification dated 22nd July 2022⁸ had been adopted by AERC. The MoP had revised the RPO trajectory again in October 2023⁹ which is expected to be adopted by AERC in line with its previous approach. The RPO Trajectory would require sourcing more than 43% of the total energy supply for APDCL through RE sources by FY30.

Figure 10: RPO Trajectory¹⁰



4.2 Renewable Energy Potential in Assam

As per MNRE, Assam has a potential of around 14.24 GW of possible RE implementation in state. Out of this, solar potential makes up for 13.76 GW. However, with respect to actual implementation, only 248 MW solar capacity addition has been achieved till date. The other RE implemented has been hydro including SHP (42.6 MW).

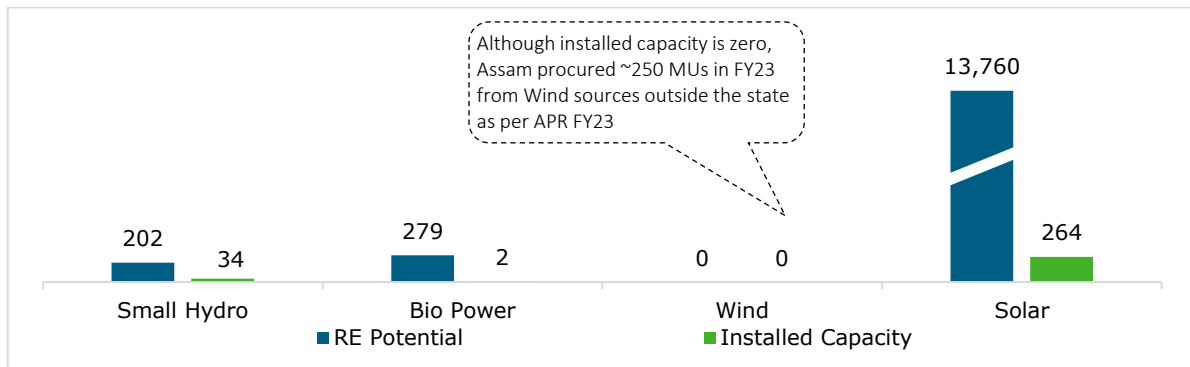
⁸ Source:

https://powermin.gov.in/sites/default/files/Renewable_Purchase_Obligation_and_Energy_Storage_Obligation_Trajectory_till_2029_30.pdf;
<https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/10/202310051180404775.pdf>

⁹ Source: https://powermin.gov.in/sites/default/files/webform/notices/Notification_Regarding_Renewable_Purchase_Obligation_RPO.pdf

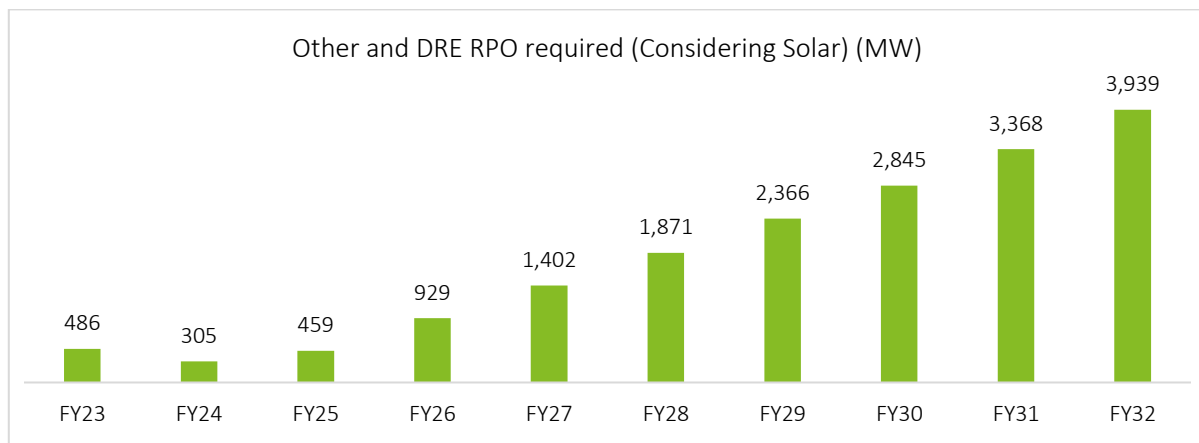
¹⁰ Note: The RPO target for FY31 and FY32 has been extrapolated basis incremental difference of RPO targets between FY29 and FY30.

Figure 11: RE Potential vs Installed Capacity (MW) in Assam



With limited potential from small hydro projects (SHP), large hydro, wind and biomass in the state, Solar based capacity expansion is an option for achieving the RPO targets and lower the APPC for Assam. Based on the planned capacity expansion for Wind and Hydro based generation / power procurement, Assam would need about 3,939 MW of solar capacity to meet its Obligation for meeting Distributed Renewable Energy (DRE) and Other RPO by FY 2032. The State has only 250 MW of Solar capacity currently leading to deficit of about ~500 MW capacity to begin with in FY23 itself¹¹.

Figure 12: Other and DRE RPO Capacity Requirement in Assam (MW; Expected from Solar)



As can be seen from the graph, above, APDCL may require about ~1000 MW of additional Solar capacity in its portfolio from FY26 itself. Thus, it is imperative to plan and setup large solar projects in the State to mitigate the RPO deficit.

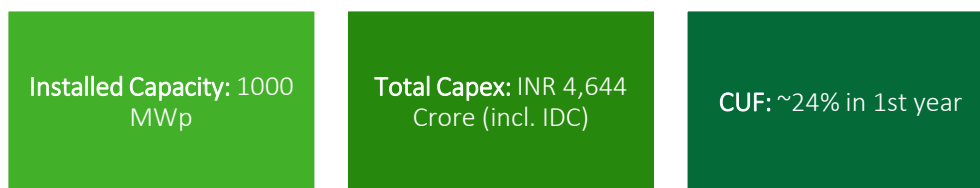
¹¹ Source: Deloitte Analysis; Assumed Other + DRE RPO to be met by Solar Capacity at average CUF of ~19% (as per AERC Norms for CUF for Solar Projects under RE Tariff Regulations, 2017); Other RPO Trajectory for FY31 and FY32 extrapolated based on incremental RPO between FY29 and FY30 as set by MoP/AERC

5 Impact of proposed 1000 MWp Solar Project in Assam on the Average Power Procurement Cost for APDCL

The proposed 1000 MWp Solar Project has the potential to significantly reduce APPC for Assam. The project size will also help APDCL to rapidly catchup with RPO Targets in the State and give the requisite push for setting up of large-scale projects in the State. Some of the key project features¹² are enumerated below:

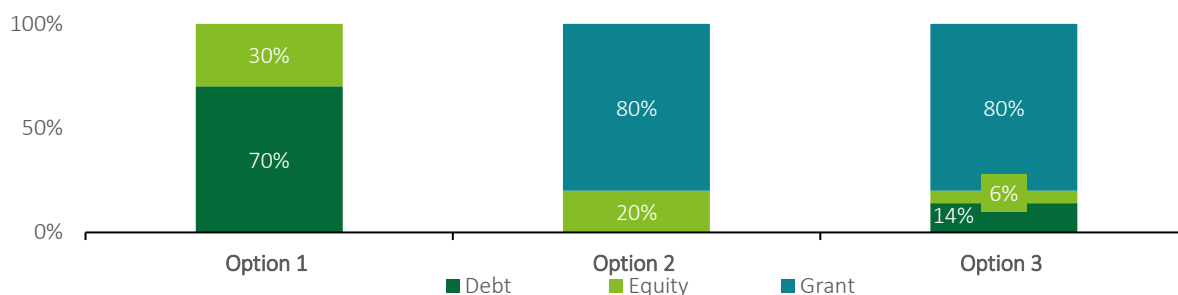
5.1 Project Details and LCOE projections

Figure 13: Key Project Details of Proposed 1000 MWp Solar Project



The project can be funded under multiple options through Debt/Equity/Grant. The proposed options for the project can be:

Figure 14: Proposed funding Options

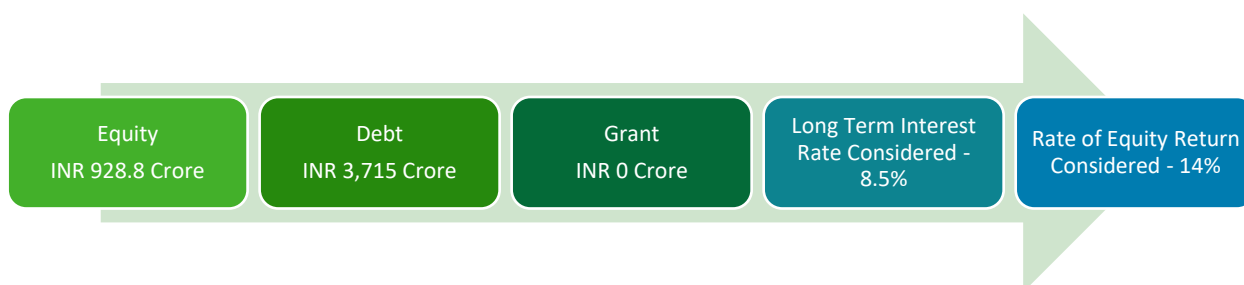


We have analyzed the project's LCOE in each of the above funding options below:

5.1.1 Scenario 1: LCOE for Debt-Equity Ratio of 70:30

The key financial parameters for the project under Option 1 has been assumed to be as follows:

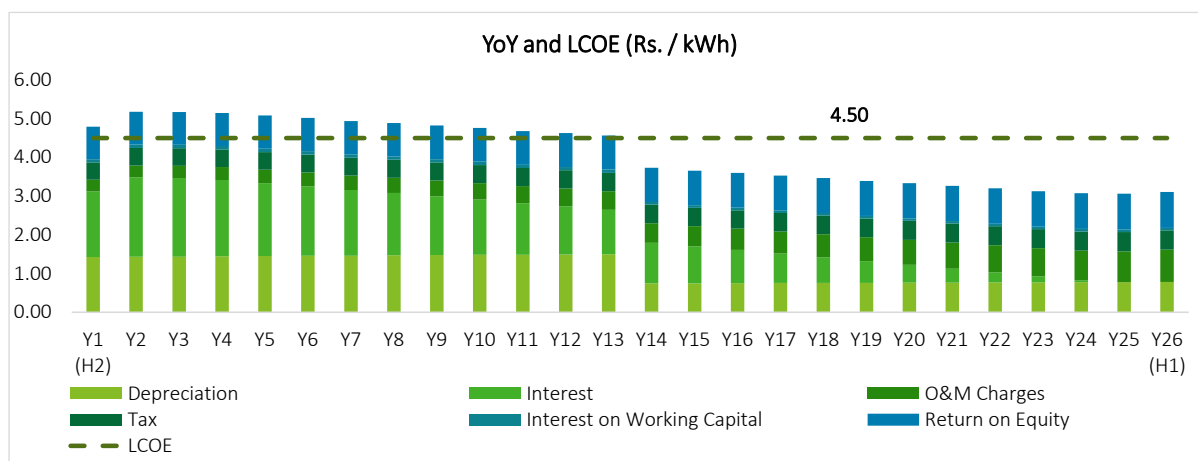
Figure 15: Key Financial Parameters for Project Funding under Option 1



The detailed assumptions have been enumerated in Annexure I. The project's LCOE is calculated to be Rs. ~4.50 / kWh

¹² Source: Draft DPR; APDCL

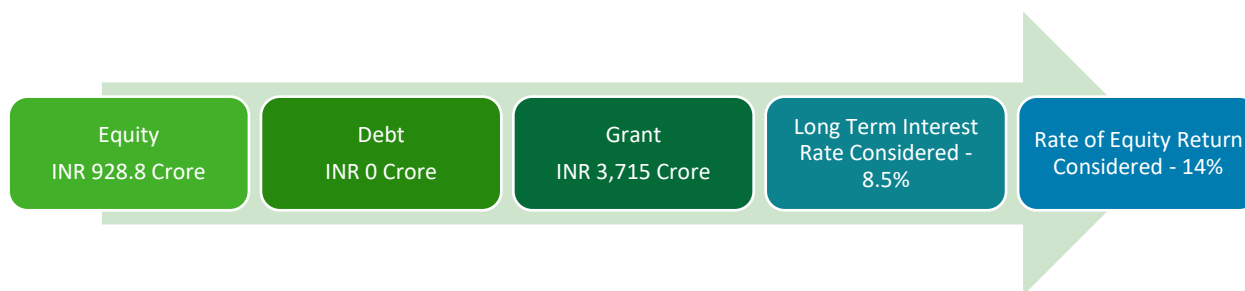
Figure 16: YoY and LCOE under Option 1



5.1.2 Scenario 2: LCOE for Grant: Equity Ratio of 80:20

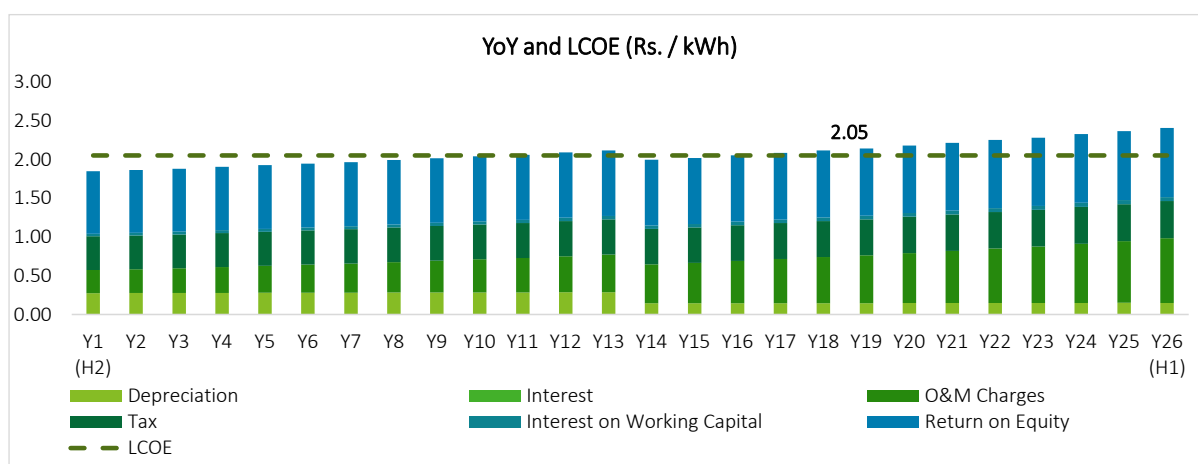
The key financial parameters for the project under Option 2 has been assumed to be as follows:

Figure 17: Key Financial Parameters for Project Funding under Option 2



The detailed assumptions have been enumerated in Annexure II. The project LCOE is computed to be Rs. 2.05 / kWh

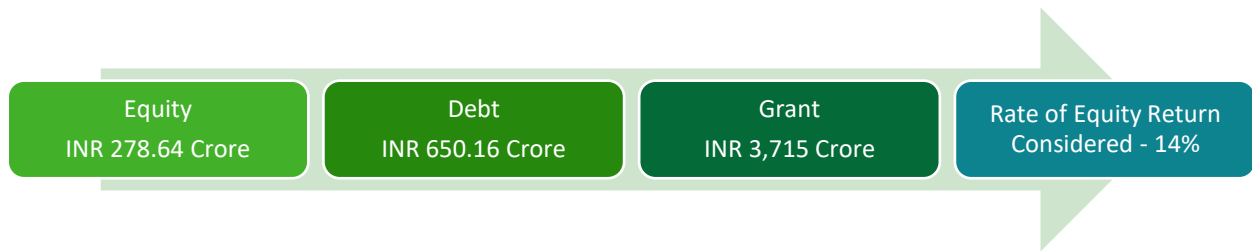
Figure 18: YoY and LCOE under Option 2



5.1.3 Scenario 3: LCOE for Debt-Equity-Grant ratio of 14:6:80

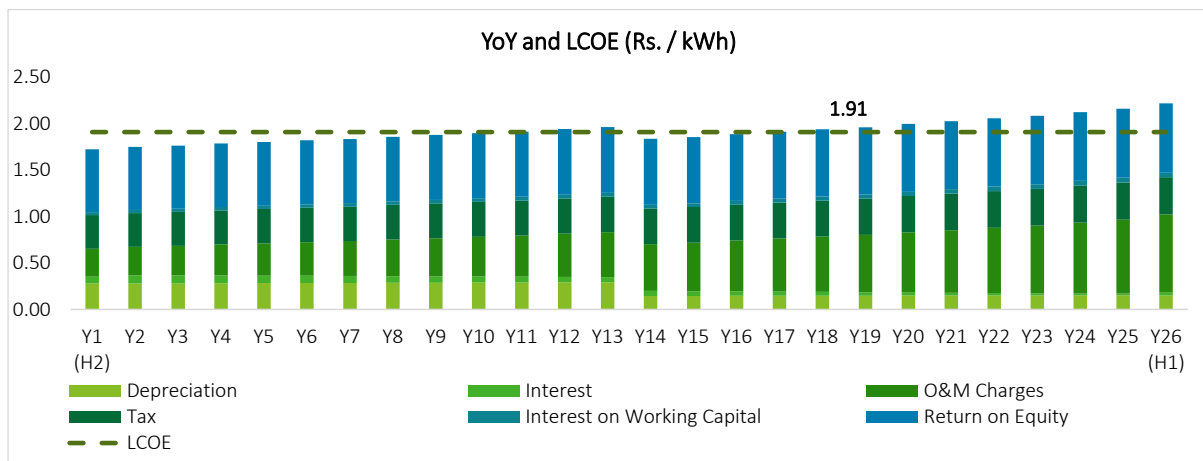
This option has been considered basis AERC's historical approach towards grant funded projects in the State. In this option, AERC is expected to consider normative Equity and Debt Structure after reduction of Grant Component from the Capex for the project. The key financial parameters for the project under Option 3 has been assumed to be as follows:

Figure 19: Key Financial Parameters for Project Funding under Option 3



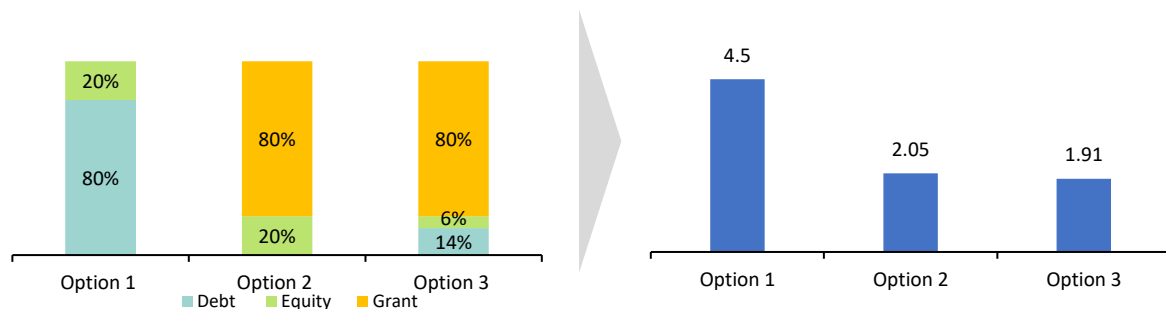
The detailed assumptions have been enumerated in Annexure III. The project LCOE is computed to be Rs. 1.91 / kWh

Figure 20: YoY and LCOE under Option 3



Thus, under the three options, the following LCOE are expected for the 1000 MWp Solar Project:

Figure 21: Proposed Funding Options and LCOE projections respectively



As can be seen, under all 3 options, the LCOE of the project will be significantly less than the current APPC of Rs. ~5.4 / kWh for APDCL. The project can significantly reduce the power purchase cost burden for APDCL in future and reduce the ACOS-ABR Gap.

5.2 Impact of the project on Average Power Procurement Costs for APDCL

The impact of the proposed project on APDCL's Power Procurement Cost (PPC) is expected to be as follows:

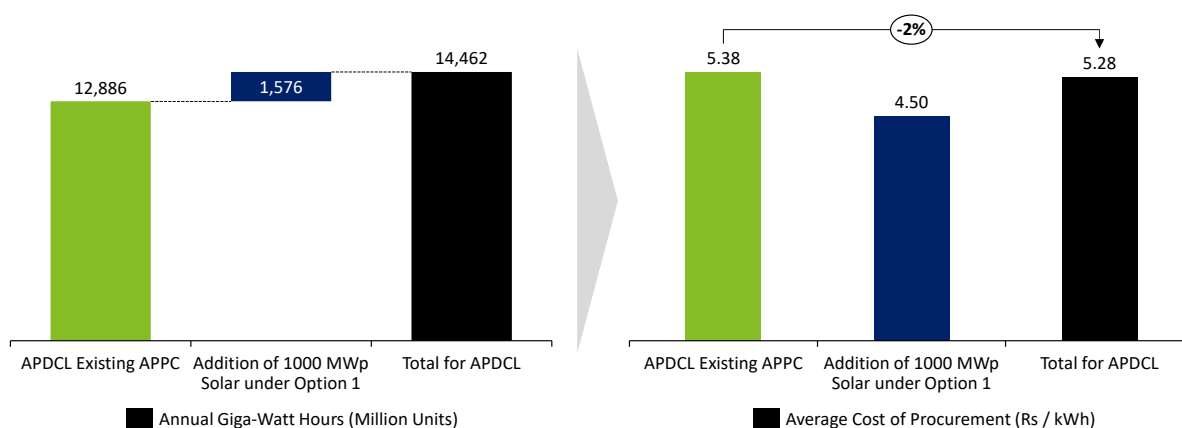
Table 3: Power Purchase Cost Analysis for APDCL

Tariff Analysis	Existing for APDCL	Addition of 1000 MWp in APDCL PPC		
Particulars	FY23 (Estimated as per APR)	Option 1	Option 2	Option 3
	A	B	C	D
Total Electricity Purchased (MUs; Annual)	12,886	1576	1576	1576
Cost in Rs. Crore	6,931	709	323	301
Per Unit Cost (Rs. / kWh)	5.38	4.50	2.05	1.91

The impact of the 1000 MWp in APDCL's APPC would be as follows:

5.2.1 APDCL's APPC under Option 1 Model for Implementation of 1000 MWp Solar Project

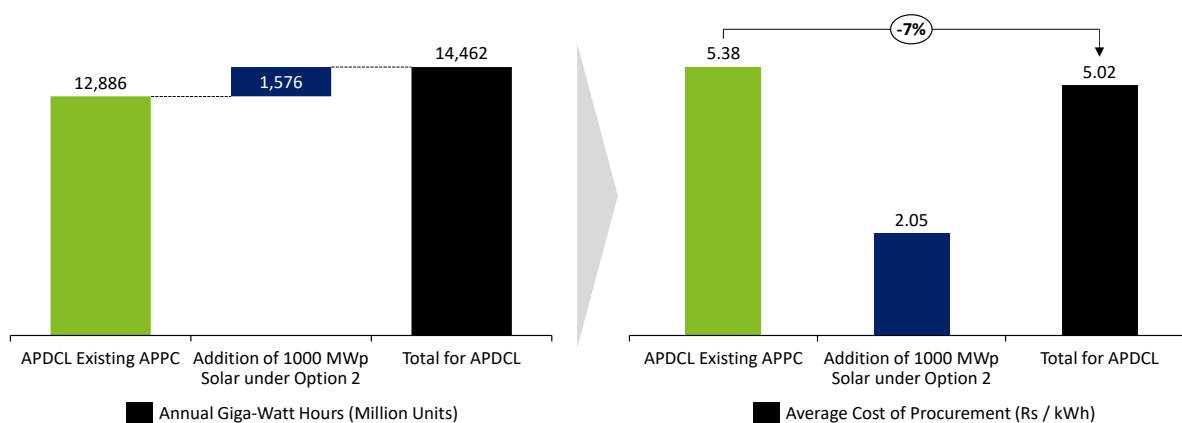
Figure 22: APPC for APDCL under Option 1 Model for Implementation of 1000 MWp Solar Project



Under Option 1, the APPC for APDCL would reduce by Rs. ~0.10 /KWh with average APPC being Rs. 5.28 / kWh.

5.2.2 APDCL's APPC under Option 2 Model for Implementation of 1000 MWp Solar Project

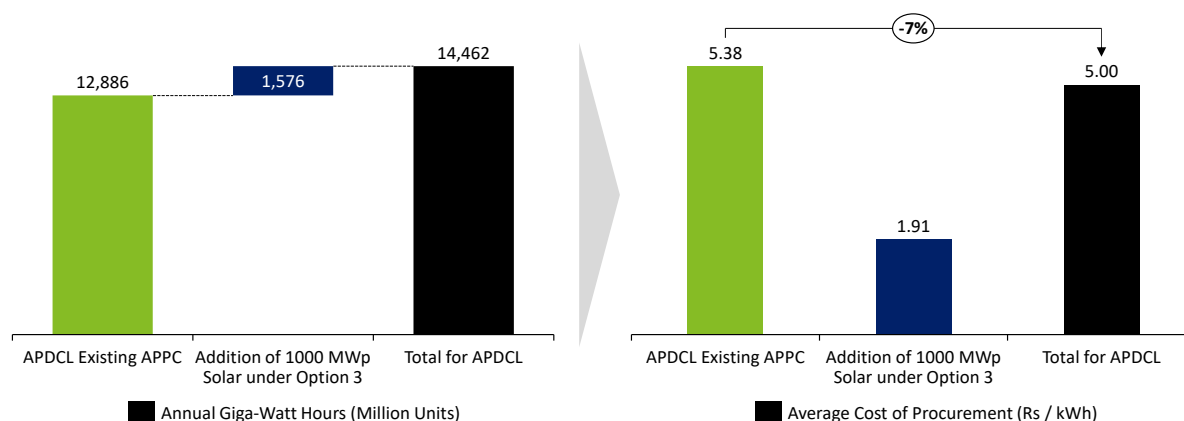
Figure 23: APPC for APDCL under Option 2 Model for Implementation of 1000 MWp Solar Project



Under Option 2, the APPC for APDCL would reduce by Rs. ~0.36 /KWh with average APPC being Rs. 5.02/ kWh.

5.2.3 APDCL's APPC under Option 3 Model for Implementation of 1000 MWp Solar Project

Figure 24: APPC for APDCL under Option 3 Model for Implementation of 1000 MWp Solar Project



Under Option 3, the APPC for APDCL would reduce by Rs. ~0.38 /KWh with average APPC being Rs. 5.00/ kWh.

Thus, the project is expected to bring APDCL's overall APPC by ~2-7% based on the implementation modality. Further the project benefits APDCL by helping it fulfil its RPO obligation till FY26.

6 Annexure

6.1 Annexure I

Assumptions for Option 1: 1000 MW Solar Plant with 80% Debt and 20% Equity

Table 4: Assumptions for Option 1 based LCOE Computation

S.N	Parameters	Units	Input	Comments
A.	Project Cost Input			
	Project Size-Contracted Capacity(AC)	MW	750	As per DPR
	Project Size -Installed Capacity(DC)	MWp	1,080	As per DPR
	PPA Effective Start Date	Date	1-Apr-24	Assumption
	Overloading Factor		1.4	Assumption
	Total Project Cost /MW	INR Cr/MW	6.01	As per DPR
	Total Project Cost	INR Cr	4,510	
	Construction Start Date	Date	1-Apr-24	Assumption
	Construction period	Years	1.5	Assumption
	Project COD	Date	1-Oct-26	Assumption
	Project Life	Years	25	As per Norms
B.	Finance Cost			
	Debt:Equity:Grant Ratio	x:1 ratio	80:20:0	As per Option1 Scenario
	Interest cost till COD	%	8.50%	As per Corporate Loan Rate
	Loan Tenor	Years	25	Assumption
	Repayment period	years	21.5	Assumption
	Moratorium Period (post COD)	years	2	Assumption
	Upfront Equity Requirement as % of total equity	%	30%	Assumption
	Cost of Equity	%	14.0%	As per AERC norms
	WACC	%	7.22%	
	Capex Phasing	%	FY25-66.66%, FY26-:33.33%	Assumption
C.	Capacity Utilization			
	CUF - first year (Based on Contracted Capacity)	%	24.08%	As per DPR
	Auxiliary & Transmission loss	%	0.0%	Assumption
	Degradation Annually	%	0.4%	As per DPR
D.	O & M Cost			
	Site O&M expenses	INR lakh/mw	6.00	As per DPR
	Site O&M Expenses escalation	%	3.84%	As per DPR
	Water Charges	INR Cr.	0.8	As per DPR
	Water Charges Escalation	%	1%	As per DPR
E.	Working Capital Requirement			
	Receivables – tariff	Months	2.0	As per AERC norms and DPR
	O&M Expenses	Months	1	As per AERC norms
	Maintenance spares	of O & M	15%	As per AERC norms

S.N	Parameters	Units	Input	Comments
	WC Interest cost	%	9.50%	Assumption; As per AERC it would be 11.50%
F.	Taxation			
	Income tax rate	%	34.94%	As per DPR
G.	Depreciation			
	Max permissible depreciation	%	90%	As per AERC norms
	Grant as percentage of project cost	%	0%	As per Project Plan
	Book Depreciation Rate - SLM	%	5.28%	As per AERC norms

6.2 Annexure II

Assumptions for Option 2: 1000 MW Solar Plant with 20% Equity and 80% Grant

Table 5: Assumptions for Option 2 based LCOE Computation

S.N	Parameters	Units	Input	Comments
A.	Project Cost Input			
	Project Size-Contracted Capacity(AC)	MW	750	As per DPR
	Project Size -Installed Capacity(DC)	MWp	1,080	As per DPR
	PPA Effective Start Date	Date	1-Apr-24	Assumption
	Overloading Factor		1.4	Assumption
	Total Project Cost /MW	INR Cr/MW	6.01	As per DPR
	Total Project Cost	INR Cr	4,510	
	Construction Start Date	Date	1-Apr-24	Assumption
	Construction period	Years	1.5	Assumption
	Project COD	Date	1-Oct-26	Assumption
	Project Life	Years	25	As per Norms
B.	Finance Cost			
	Debt:Equity:Grant Ratio	x:1 ratio	0:20:80	As per Option2 Scenario
	Interest cost till COD	%	8.50%	As per Corporate Loan Rate
	Loan Tenor	Years	25	Assumption
	Repayment period	years	21.5	Assumption
	Moratorium Period (post COD)	years	2	Assumption
	Upfront Equity Requirement as % of total equity	%	30%	Assumption
	Cost of Equity	%	14.0%	As per AERC norms
	WACC	%	2.80%	
	Capex Phasing	%	FY25-66.66%, FY26-:33.33%	Assumption
C.	Capacity Utilization			
	CUF - first year (Based on Contracted Capacity)	%	24.08%	As per DPR
	Auxiliary & Transmission loss	%	0.0%	Assumption
	Degradation Annually	%	0.4%	As per DPR
D.	O & M Cost			
	Site O&M expenses	INR lakh/mw	6.00	As per DPR
	Site O&M Expenses escalation	%	3.84%	As per DPR

S.N	Parameters	Units	Input	Comments
	Water Charges	INR Cr.	0.8	As per DPR
	Water Charges Escalation	%	1%	As per DPR
E.	Working Capital Requirement			
	Receivables – tariff	Months	2.0	As per AERC norms and DPR
	O&M Expenses	Months	1	As per AERC norms
	Maintenance spares	of O & M	15%	As per AERC norms
	WC Interest cost	%	9.50%	Assumption; As per AERC it would be 11.50%
F.	Taxation			
	Income tax rate	%	34.94%	As per DPR
G.	Depreciation			
	Max permissible depreciation	%	90%	As per AERC norms
	Grant as percentage of project cost	%	0%	As per Project Plan
	Book Depreciation Rate - SLM	%	5.28%	As per AERC norms

6.3 Annexure III

Assumptions for Option 3: 1000 MW Solar Plant with 14% Debt, 6% Equity and 80% Grant

Table 6: Assumptions for Option 3 based LCOE Computation

S.N	Parameters	Units	Input	Comments
A.	Project Cost Input			
	Project Size-Contracted Capacity(AC)	MW	750	As per DPR
	Project Size -Installed Capacity(DC)	MWp	1,080	As per DPR
	PPA Effective Start Date	Date	1-Apr-24	Assumption
	Overloading Factor		1.4	Assumption
	Total Project Cost /MW	INR Cr/MW	6.01	As per DPR
	Total Project Cost	INR Cr	4,510	
	Construction Start Date	Date	1-Apr-24	Assumption
	Construction period	Years	1.5	Assumption
	Project COD	Date	1-Oct-26	Assumption
	Project Life	Years	25	As per Norms
B.	Finance Cost			
	Debt:Equity:Grant Ratio	x:1 ratio	14:6:80	As per Option3 Scenario
	Interest cost till COD	%	8.50%	As per Corporate Loan Rate
	Loan Tenor	Years	25	Assumption
	Repayment period	years	21.5	Assumption
	Moratorium Period (post COD)	years	2	Assumption
	Upfront Equity Requirement as % of total equity	%	30%	Assumption
	Cost of Equity	%	14.0%	As per AERC norms
	WACC	%	1.61%	
	Capex Phasing	%	FY25-66.66%, FY26-:33.33%	Assumption
C.	Capacity Utilization			

S.N	Parameters	Units	Input	Comments
	CUF - first year (Based on Contracted Capacity)	%	24.08%	As per DPR
	Auxiliary & Transmission loss	%	0.0%	Assumption
	Degradation Annually	%	0.4%	As per DPR
D.	O & M Cost			
	Site O&M expenses	INR lakh/mw	6.00	As per DPR
	Site O&M Expenses escalation	%	3.84%	As per DPR
	Water Charges	INR Cr.	0.8	As per DPR
	Water Charges Escalation	%	1%	As per DPR
E.	Working Capital Requirement			
	Receivables – tariff	Months	2.0	As per AERC norms and DPR
	O&M Expenses	Months	1	As per AERC norms
	Maintenance spares	of O & M	15%	As per AERC norms
	WC Interest cost	%	9.50%	Assumption; As per AERC it would be 11.50%
F.	Taxation			
	Income tax rate	%	34.94%	As per DPR
G.	Depreciation			
	Max permissible depreciation	%	90%	As per AERC norms
	Grant as percentage of project cost	%	0%	As per Project Plan
	Book Depreciation Rate - SLM	%	5.28%	As per AERC norms



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